HSC Mathematics Extension 1 & 2: Trigonometry Mastery

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"Everything that has a beginning has an end."

The Matrix

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1 Introduction

1.1 Project Overview

This booklet is a growing collection of trigonometry problems for NSW HSC Mathematics Extension 1 and Extension 2 students. It covers the full trigonometry strand across both syllabuses, including right-angled triangles, trigonometric functions of a general angle, trigonometric identities and equations, the sine and cosine rules, inverse trigonometric functions, compound and double-angle formulae, the t-formulae, and products-to-sums conversions. The problems range from straightforward skill-drill questions to multi-step challenge problems suitable for exam preparation and beyond.

1.2 Target Audience

This resource is written for students who want focused, exam-oriented practice with trigonometry in a clear, classroom-friendly format. Solutions are intentionally concise but each one highlights the key idea behind the answer. Students who work systematically through Parts 1 and 2 should gain both fluency and a deeper conceptual understanding of the trigonometry topics tested in Extension 1 and Extension 2.

1.3 How to Use This Booklet

- Read the core formulas and theory summaries in this introduction first.
- Attempt each problem before reading the solution.
- Pay attention to the *type* of question: is it an identity, an equation, a geometry application, or an inverse trigonometry question?
- Look for structure: symmetry, compound-angle patterns, Pythagorean identities, or opportunities to substitute $t = \tan(\theta/2)$.
- Use the remarks and takeaways to build pattern recognition for future HSC questions.

1.4 Extension 1 Knowledge Needed

Most problems in this booklet can be solved using standard HSC Mathematics Extension 1 and Extension 2 knowledge. The key background material is:

- **Right-angled trigonometry:** definitions of $\sin\theta$, $\cos\theta$, $\tan\theta$ as ratios of opposite, adjacent, and hypotenuse; exact values for $30°$, $45°$, $60°$.
- **ASTC rule (quadrant signs):** All, Sine, Tangent, Cosine are positive in quadrants I, II, III, IV respectively.
- **Trigonometric functions of a general angle:** converting angles to the related acute angle and applying the correct sign.
- **Fundamental identity:** $\sin^2\theta + \cos^2\theta = 1$ and its derived forms $1 + \tan^2\theta = \sec^2\theta$ and $1 + \cot^2\theta = \csc^2\theta$.
- **Double and symmetry identities:** supplementary $(\pi - \theta)$, complementary $(\pi/2 - \theta)$, and reflex $(2\pi - \theta)$ angle reductions.

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- **The sine rule and cosine rule:** applied to general (non-right-angled) triangles, including the ambiguous case.
- **Area formula:** Area $= \frac{1}{2}ab\sin C$.
- **Inverse trigonometric functions:** restricted domains for arcsin, arccos, arctan and graphical properties.
- **Compound-angle identities:** $\sin(\alpha \pm \beta)$, $\cos(\alpha \pm \beta)$, $\tan(\alpha \pm \beta)$.
- **Double-angle formulae:** $\sin 2\theta$, $\cos 2\theta$ (all three forms), $\tan 2\theta$.
- **The t-formulae:** expressing $\sin\theta$, $\cos\theta$, $\tan\theta$ in terms of $t = \tan(\theta/2)$.
- **Products-to-sums and sums-to-products:** converting between products and sums of trigonometric functions.

1.5 Core Formulas and Ideas

Exact Values.

θ	$\sin\theta$	$\cos\theta$	$\tan\theta$
$0°$	0	1	0
$30°$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
$45°$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	1
$60°$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
$90°$	1	0	undefined

Pythagorean Identities.

$$\sin^2\theta + \cos^2\theta = 1, \qquad 1 + \tan^2\theta = \sec^2\theta, \qquad 1 + \cot^2\theta = \csc^2\theta.$$

Sine Rule.

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R,$$

where R is the circumradius of the triangle.

Cosine Rule.

$$c^2 = a^2 + b^2 - 2ab\cos C, \qquad \cos C = \frac{a^2 + b^2 - c^2}{2ab}.$$

Area Formula.

$$\text{Area} = \tfrac{1}{2}ab\sin C.$$

1.6 Angle Reduction Identities

For all angles θ:

$$\sin(\pi - \theta) = \sin\theta, \qquad \cos(\pi - \theta) = -\cos\theta, \qquad \tan(\pi - \theta) = -\tan\theta.$$

$$\sin(\pi + \theta) = -\sin\theta, \qquad \cos(\pi + \theta) = -\cos\theta, \qquad \tan(\pi + \theta) = \tan\theta.$$

$$\sin(2\pi - \theta) = -\sin\theta, \qquad \cos(2\pi - \theta) = \cos\theta.$$

$$\sin\left(\tfrac{\pi}{2} - \theta\right) = \cos\theta, \qquad \cos\left(\tfrac{\pi}{2} - \theta\right) = \sin\theta.$$

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1.7 Compound Angle Identities

$$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta.$$
$$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta.$$
$$\tan(\alpha \pm \beta) = \frac{\tan\alpha \pm \tan\beta}{1 \mp \tan\alpha\tan\beta}.$$

1.8 Double-Angle Formulae

$$\sin 2\theta = 2\sin\theta\cos\theta.$$
$$\cos 2\theta = \cos^2\theta - \sin^2\theta = 1 - 2\sin^2\theta = 2\cos^2\theta - 1.$$
$$\tan 2\theta = \frac{2\tan\theta}{1 - \tan^2\theta}.$$

The power-reduction forms (useful for integration in Extension 2):

$$\sin^2\theta = \frac{1 - \cos 2\theta}{2}, \qquad \cos^2\theta = \frac{1 + \cos 2\theta}{2}.$$

1.9 The t-Formulae

Let $t = \tan\left(\frac{\theta}{2}\right)$. Then:

$$\sin\theta = \frac{2t}{1+t^2}, \qquad \cos\theta = \frac{1-t^2}{1+t^2}, \qquad \tan\theta = \frac{2t}{1-t^2}.$$

These substitutions convert trigonometric equations into polynomial equations in t, a powerful technique for solving equations of the form $a\sin\theta + b\cos\theta = c$.

Remark 1.1

The t-substitution introduces the extraneous restriction $\theta \neq \pi + 2k\pi$ (where $\tan(\theta/2)$ is undefined). Always check that solutions from the t-substitution are valid in the original equation, and consider separately whether $\theta = \pi$ is a solution.

1.10 Products-to-Sums and Sums-to-Products

Products to sums:

$$\sin\alpha\cos\beta = \tfrac{1}{2}[\sin(\alpha+\beta) + \sin(\alpha-\beta)].$$
$$\cos\alpha\cos\beta = \tfrac{1}{2}[\cos(\alpha-\beta) + \cos(\alpha+\beta)].$$
$$\sin\alpha\sin\beta = \tfrac{1}{2}[\cos(\alpha-\beta) - \cos(\alpha+\beta)].$$

Sums to products:

$$\sin A + \sin B = 2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right).$$
$$\sin A - \sin B = 2\cos\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right).$$
$$\cos A + \cos B = 2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right).$$
$$\cos A - \cos B = -2\sin\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right).$$

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1.11 Inverse Trigonometric Functions

The three principal inverse functions and their restricted domains are:

$$\arcsin : [-1,1] \to \left[-\tfrac{\pi}{2}, \tfrac{\pi}{2}\right], \qquad \arccos : [-1,1] \to [0,\pi], \qquad \arctan : \mathbb{R} \to \left(-\tfrac{\pi}{2}, \tfrac{\pi}{2}\right).$$

Key identities:

$$\arcsin(-x) = -\arcsin x, \qquad \arccos(-x) = \pi - \arccos x, \qquad \arctan(-x) = -\arctan x.$$

$$\arcsin x + \arccos x = \frac{\pi}{2} \quad \text{for } x \in [-1,1].$$

1.12 Auxiliary Angle Form

Any expression of the form $a\sin\theta + b\cos\theta$ can be written as a single sinusoid:

$$a\sin\theta + b\cos\theta = R\sin(\theta + \varphi),$$

where

$$R = \sqrt{a^2+b^2}, \qquad \tan\varphi = \frac{b}{a}.$$

Similarly, $a\sin\theta + b\cos\theta = R\cos(\theta - \psi)$ where $\tan\psi = a/b$.

This form is extremely useful for finding the maximum and minimum values of weighted combinations of sine and cosine, and for solving equations of the form $a\sin\theta + b\cos\theta = c$.

1.13 Three-Dimensional Trigonometry

Many HSC problems extend two-dimensional trigonometry to three-dimensional figures (pyramids, prisms, slanted poles, etc.). A reliable method is:

1. Draw a clear diagram and label all known lengths and angles.
2. Identify the right-angled triangles embedded in the figure.
3. Work systematically from known information, using the sine rule or cosine rule where required.
4. Use the angle of elevation or depression carefully: it is always measured from the horizontal.

1.14 Problem-Solving Habits for This Booklet

- Always write down the domain of θ before solving a trigonometric equation.
- Check whether the identity or formula requires degrees or radians.
- For the sine rule, always check the ambiguous case when the given angle is acute.
- In identity proofs, start from the more complex side and aim to simplify.
- For inverse trigonometric function questions, sketch the graph to confirm the range.
- In t-substitution problems, remember to check $\theta = \pi$ separately.

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Remark 1.2

The most common trigonometry mistake is ignoring the full solution set when solving equations. For example, $\sin\theta = \frac{1}{2}$ has infinitely many solutions; always write the general solution and then restrict to the given domain.

Remark 1.3 (Why trigonometry matters beyond the HSC)

Trigonometric functions appear throughout mathematics, physics, engineering, and data science. Fourier analysis decomposes any periodic signal – sound, images, electrical waveforms – into sums of sines and cosines. The t-substitution and product-to-sum identities that appear in HSC Extension 2 are the same tools used in signal processing, control theory, and quantum mechanics.

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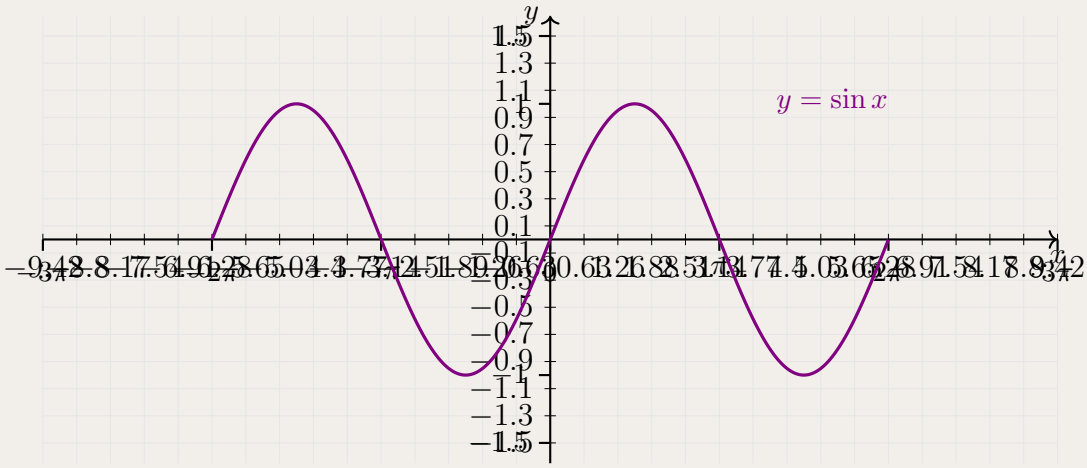
2 Part 1: Worked Problems

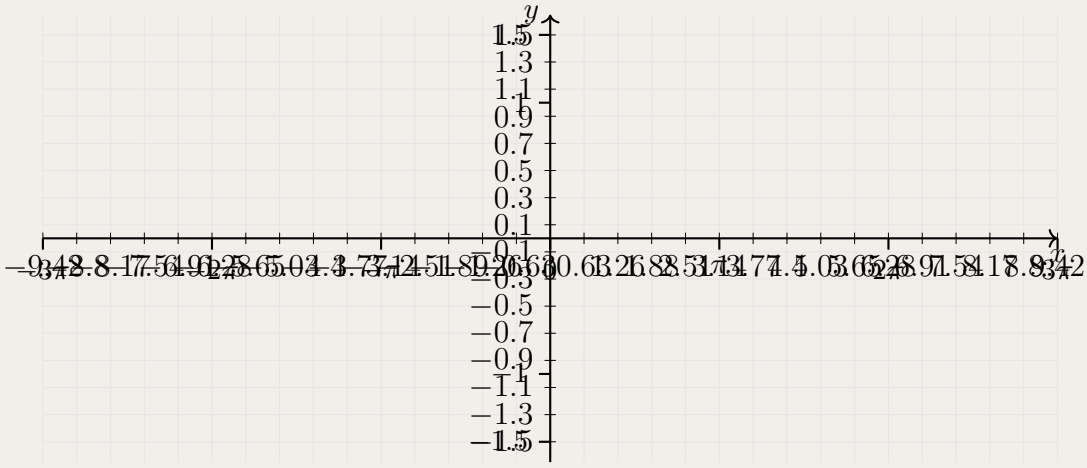
2.1 Basic

Problem 2.1: Graphing a Sine Function and its Phase Shift

The graph of $y = \sin x$ (drawn in purple) is shown below for $x \in [-2\pi,\, 2\pi]$.



On the axes below, draw the graph of $y = \sin\left(x - \dfrac{\pi}{2}\right)$ for $x \in [-2\pi,\, 2\pi]$.



Solution 2.1

The graph of $y = \sin\left(x - \frac{\pi}{2}\right)$ is the graph of $y = \sin x$ shifted **right** by $\frac{\pi}{2}$ (a phase shift of $\frac{\pi}{2}$).

Using the compound-angle identity:

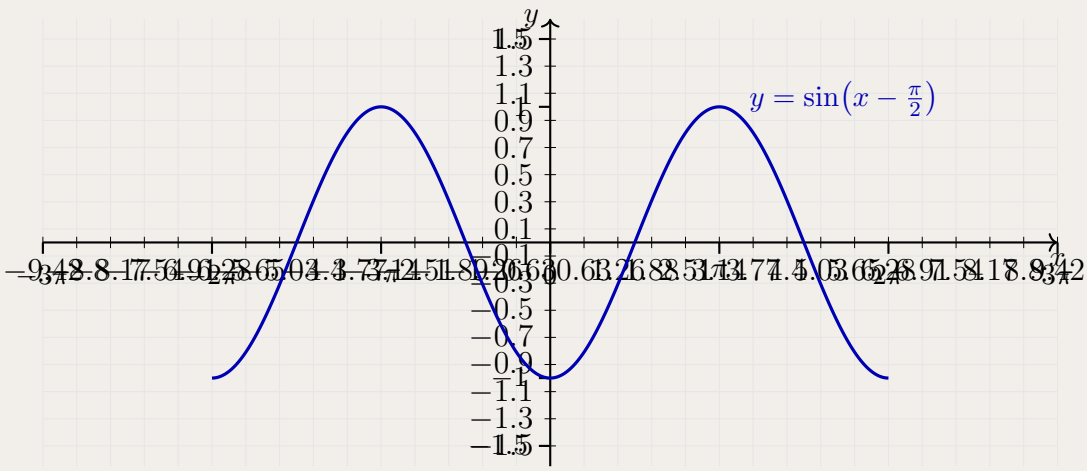
$$\sin\left(x - \frac{\pi}{2}\right) = \sin x \cos\frac{\pi}{2} - \cos x \sin\frac{\pi}{2} = \sin x \cdot 0 - \cos x \cdot 1 = -\cos x.$$

Hence $y = \sin\left(x - \frac{\pi}{2}\right) = -\cos x$.

Key points on $[-2\pi, 2\pi]$:

x	-2π	$-\frac{3\pi}{2}$	$-\pi$	$-\frac{\pi}{2}$	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
y	-1	0	1	0	-1	0	1	0	-1

The completed graph (curve shown in blue):



Takeaways 2.1

- **Phase shift:** The graph of $y = \sin(x - c)$ for $c > 0$ is the graph of $y = \sin x$ shifted c units to the *right*.
- **Identity connection:** $\sin\left(x - \frac{\pi}{2}\right) = -\cos x$, so this curve is also the reflection of $y = \cos x$ in the x-axis.
- **Graphical check:** The shifted sine reaches its zeros at $x = \frac{\pi}{2} + k\pi$ ($k \in \mathbb{Z}$), its maximum of 1 at $x = \pi + 2k\pi$, and its minimum of -1 at $x = 2k\pi$ — each exactly $\frac{\pi}{2}$ to the right of the corresponding feature on $y = \sin x$.

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2.2 Medium

2.3 Advanced

3 Part 2: Practice and Challenge Bank

3.1 Warm-Up Drills

3.2 Stretch Problems

3.3 Challenge Corner

4 Conclusion

Trigonometry rewards careful attention to detail as much as formula recall. Many HSC questions become far more tractable once you identify the correct strategy: choosing the right form of $\cos 2\theta$, recognising a products-to-sums pattern, or knowing instinctively when the t-substitution will linearise an equation. As this booklet grows, it will serve as a compact reference for the most important trigonometric patterns students encounter in Extension 1 and Extension 2 examinations and beyond.

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A Appendices

A.1 The Signs of the Trigonometric Functions

The signs of the trigonometric functions depend only on the signs of x and y. (The radius r is always positive.) The signs of x and y depend in turn only on the quadrant in which the ray lies. The table below summarises the sign of each function in each quadrant; the accompanying diagram shows the coordinate signs on the unit circle.

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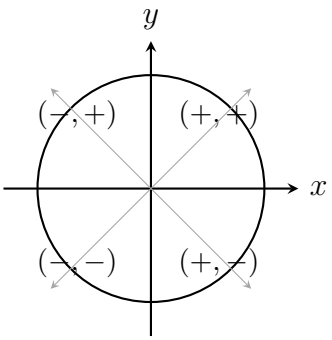
Function	**1st**	**2nd**	**3rd**	**4th**
x	$+$	$-$	$-$	$+$
y	$+$	$+$	$-$	$-$
r	$+$	$+$	$+$	$+$
$\sin\theta$	$+$	$+$	$-$	$-$
$\cos\theta$	$+$	$-$	$-$	$+$
$\tan\theta$	$+$	$-$	$+$	$-$
$\csc\theta$	$+$	$+$	$-$	$-$
$\sec\theta$	$+$	$-$	$-$	$+$
$\cot\theta$	$+$	$-$	$+$	$-$



$\csc\theta$ has the same sign as $\sin\theta$; $\sec\theta$ has the same sign as $\cos\theta$; $\cot\theta$ has the same sign as $\tan\theta$.

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